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1. ■■ Why NETRA? — Indian Cyber Threat Reality

India mein deepfake-based cybercrimes ek national crisis ban chuka hai. Teen major scam categories:

- Digital Arrest Scam:** Fake police officer ki deepfake video call par retired uncle ko 2 lakh रुपये देने पर मजबूर. Sirf 2024 mein Rs 120.30 crore aise scams mein gaye.
- Stock Market VIP Scam:** Mukesh Ambani, Adani ya Sudha Murty ki fake videos banakar Telegram groups mein '500% guaranteed returns' promote kiye jate hain.
- Voice Clone Kidnapping:** Parents ko unke bache ki bilkul asli cloned awaaz mein call: 'Mujhe bachao!' — Real-time emotional manipulation.

Western deepfake detectors Indian faces par kaam nahi karte — unhe Indian demographics, skin tones, aur regional video quality par train hi nahi kiya gaya. NETRA yeh gap bridge karta hai.

- **NETRA Ka Mission**

Teesri Aankh (Third Eye) — har photo, video, aur audio ko seconds mein scan karke court-ready mathematical proof ke saath verdict dena.

100% Indian demographics ke liye optimized. Zero hallucinations. Zero fake promises.

2. ■■ Architecture Overview — What We Actually Built

NETRA ek multi-modal, multi-layer forensic engine hai jo 5 parallel detection channels run karta hai:

Detection Channel	Technology	What It Detects	Latency
Visual Spatial	EfficientNet-B4 + SBI Training	Face swap boundaries, blending seams	~25ms
Foundation Visual	GenD ViT-L/14 (WACV 2026)	Cross-domain deepfakes, GAN artifacts	~110ms
CLIP Probe	CLIP ViT-L/14 Linear Probe	Semantic image-text inconsistency	~40ms
Audio Deepfake	Wav2Vec2 Pretrained	TTS voice cloning, vocoder artifacts	~60ms
Physics Engine	4-Pillar Biometric System	Corneal parallax, rPPG pulse, lip sync	~80ms

Ye sab scores GatedFusionEngine mein combine hote hain — rule-based weighted fusion jo final verdict deta hai.

- **Fusion Engine:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/fusion.py>
- **GenD Engine:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/gend_engine.py
- **Spatial Detector:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/detectors/spatial.py>
- **Audio Detector:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/detectors/audio.py>

3. ■ The Pivot — Why We Moved Away From LLM-Based Detection

Initial idea: basic detector + LLM (Claude/GPT/Bedrock) se report generate karna. Hum is approach se poori tarah door chale gaye:

Problem With LLM Approach	Why It's Fatal For Forensics
LLM video/image ke pixels nahi dekhta	Detector score galat ho to LLM confident jhooth (hallucination) likhega
Court inadmissible	Indian Evidence Act mein chatbot output proof nahi hota — physical mathematical proof chahiye
3-6 second latency + API cost	Real-time scanning impossible; production mein har call ka paisa lagta hai
Not Indian-specific	Western LLMs Indian face nuances, skin tones, regional languages nahi samajhte

■ **Our Solution: Deterministic + Physics-Based**

Pure Computer Vision + Biological Physics = zero hallucination, zero LLM dependency.

Sub-second mathematical verdict. Self-hostable. Court-ready proof.

4. ■■ Training Pipeline & Datasets

4.1 How NETRA Was Trained

NETRA ka spatial backbone Kaggle T4/P100 GPU environment par train hua. Key techniques:

SBI (Self-Blended Images): Real Indian faces ke andar hi fake blend inject karke synthetic training samples banaye — alag deepfake dataset ki zaroorat nahi.

PairedSupConLoss: Supervised contrastive loss jo model ko identity se nahi, manipulation artifacts se detect karna sikhata hai — identity shortcut prevent karta hai.

Anti-Shortcut Augmentations: WhatsApp JPEG compression (quality 30-90), motion blur, downsampling — real-world compressed videos par bhi accurate.

Differential Learning Rates: EfficientNet-B4 backbone 5x slower update, classification head 5x faster.

■ **Kaggle Training Script (Main):** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/kaggle-spatial-notebook/script.py>

■ **V2 Training Engine:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/train_netra_v2.py

■ **Augmentation Pipeline:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/augmentations.py>

■ **Dataset Builder (SBI):** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/dataset_builder.py

■ **NETRA v2 Architecture:**

https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/netra_v2.py

■ Tech Detail: LinearNormHead (L2 Normalization)

```
x_hat = x / ||x||_2 (project feature vector to unit sphere)
```

```
logit_k = cos(angle(x_hat, w_hat_k)) / tau where tau=0.07
```

Result: Sharp DSLR photo vs WhatsApp compressed video = same result. Magnitude-blind.

4.2 Dataset Catalog

Dataset	Source	Size	Role
Indian Face Dataset	Kaggle: aryankashyapnaveen/indian-face-dataset	35,000+ images	REAL training — Indian faces baseline
SBI Synthetic Fakes	Auto-generated via training script	35,000+ fakes	FAKE training — Poisson blend injection
IMFDB (Bollywood)	imfdb.org	50,000+ frames	Diverse poses, lighting, expressions
DF-Platter	Kaggle: DF-Platter dataset	~3GB frames	Cross-domain GAN artifact training

SDFVD Benchmark	Kaggle: Hemgg/SDFVD-video-dataset	106 videos (53R+53F)	EVAL ONLY — standard benchmark
100 Indian Figures	GitHub LFS: benchmark_datasets/generated_100_deepfake_videos/	100 HD MP4s, 296MB	Sovereign benchmark — NEVER in training
78 Celebrity Swaps	batch_benchmark_results/generated_swaps/	78 swapped images	Zero-shot test — NEVER in training

■ **Benchmark Datasets Folder:**

https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark_datasets/

■ **100 Deepfake Videos (Git LFS):** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark_datasets/generated_100_deepfake_videos/

■ **SDFVD Metadata:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark_datasets/SDFVD/metadata.json

■ **Zero Data Leakage Guarantee**

100 Indian benchmark videos aur 78 celebrity swaps ko KABHI BHI training mein use nahi kiya.

Ye strictly held-out zero-shot test sets hain — judges ke samne genuine generalization prove karne ke liye.

5. ■ MesoNet vs NETRA — Why Old Models Fail on Modern Deepfakes

Model	Architecture	Mean Fake Score	Accuracy	Verdict
NETRA (Ours)	EfficientNet-B4 + GenD ViT-L/14	94.6%	100/100 (100%)	■ Court-Ready
MesoInception-4	Multi-Scale Inception CNN (2018)	57.5%	Unreliable	■■ Borderline
MesoNet-4	4-Layer Compact CNN (2018)	51.0%	Coin Flip	■ Obsolete

Why MesoNet Got 51%? (Coin Toss)

MesoNet 2018 mein bana tha jab deepfakes low-quality autoencoders se bante the — blurry, pixelated. MesoNet ne unhi blurry compression artifacts ko detect karna seekha.

Hamari pipeline mein InSwapper-128 + GPEN-BFR-512 GAN use hua. GPEN ne face ko GAN se super-resolve karke photorealistic skin pores, natural eye reflections aur sharp details render kiye. MesoNet ne socha 'Itna sharp face? Pakka real hai!' — aur score 51% (random guess) par aa gaya.

■ **GPEN GAN Ka Effect on MesoNet**

InSwapper-128: Neural latent space mein identity swap → 128px low-res result.

GPEN-BFR-512: GAN super-resolution → realistic skin texture, eye sparkle, beard details.

MesoNet's 2018 frequency filters: GPEN output ko 'too sharp to be fake' samajh leta → FAILS completely.

How NETRA Detects at 99.2%?

Sub-pixel Blending Seams: Jaha swapped face body se composite hua hai, waha microscopic gradient mismatch hota hai — EfficientNet-B4 dhundhta hai.

Lighting Disparities: Face ka light source aur background ka ambient light direction subtle angle mein different hota hai — Reinhard LAB mismatch.

Hyperspherical Classification: L2-normalized features sirf structural artifacts detect karte hain, brightness/sharpness se confuse nahi hote.

■ **MesoNet Evaluation Script:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/otherdeepfakemodelthatwehavetobenchmarkagainest/evaluate_all_mesonet.py

■ **Live Benchmark Script:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark.py>

■ **Benchmark Runner:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark.sh>

6. ■ GenD Foundation Backbone — Why & How We Integrated It

NETRA v1 Ki 2 Critical Failures

■■ NETRA v1 Failure Report

Failure 1 — 67.9% False Positive: Real Indian DSLR studio photos ko model ne 100% FAKE bol diya.

Root Cause: Un-normalized linear head mein high sharpness = high feature magnitude = $P(\text{fake}) \rightarrow 100\%$ for any sharp real photo.

Failure 2 — 38.5% False Negative: GPEN deepfakes ko model REAL samajh raha tha.

Root Cause: SBI training ne Poisson blend boundary dhundhna sikhaya. GPEN ne ye boundaries smooth kar di → Model confused.

GenD (WACV 2026) — The Fix

GenD: Generalizable Deepfake Detection (Yermakov et al., WACV 2026). ViT-L/14 backbone, sirf 0.03% weights train kiye — baaki frozen.

Key Technique: L2 Hyperspherical Unit Projection — feature vector ki brightness aur sharpness ka effect ZERO kar deta hai. Ab model sirf face ki structural geometry dekhta hai.

■ Tech Detail: L2 Unit Projection (Simplified)

Feature vector $x \rightarrow \hat{x} = x / ||x||$ (normalize to unit sphere, magnitude=1)

Now: DSLR sharp photo aur WhatsApp compressed photo same 'direction' par hote hain

Only structural geometry matters – magnitude irrelevant

Metric	NETRA v1	GenD ViT-L/14	NETRA v2 (Integrated)
Indian GPEN Accuracy	46.8% (FAIL)	94.2%	95.1%
Fake Recall	61.5%	100%	99.2%
DSLR False Alarms	67.9% (Critical)	<5%	<3.5%
Per-Frame Latency	25ms	110ms	~45ms (cascaded)

Two-Stage Cascade Router

Stage 1 — NETRA Fast (25ms): 80% normal videos instantly scan. Clear verdict? Done. Ambiguous (score 0.25-0.75)? → Stage 2.

Stage 2 — GenD Foundation Arbiter (110ms): Sirf borderline cases yahan aate hain. Hyperspherical model final call leta hai.

Result: Average effective latency ~45ms per frame. 95%+ cross-domain accuracy!

■ GenD Engine (ViT-L/14): https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/gend_engine.py

■ NETRA v2 Model (LinearNormHead):

https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/netra_v2.py

■ Cascade Router: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/inference/cascade_router.py

7. ■■ The 4 Physical Pillars Engine — AI Physics Nahi Tod Sakta

Deepfake AI 2D pixel manipulation karta hai. Lekin real world mein 3D physics aur human biology ke laws hain jo AI replicate nahi kar sakta. Ye 4 pillars unhe test karte hain:

Pillar 1 — Spatial Frequency Residuals

Technique: 2D DCT (Discrete Cosine Transform) + Laplacian edge convolution. Face swap mein sub-pixel boundary energy distribution real image se match nahi karta.

■ Frequency Analyzer: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/frequency_analyzer.py

Pillar 2 — Indic Phoneme-Viseme Biomechanics

Technique: Articulatory Cross-Correlation Index (ACCI). Indian languages mein retroflex consonants (■, ■, ■, ■) bolne par lips ki speed ek specific biological pattern follow karti hai.

Wav2Lip jaise deepfake tools English phoneme priors par train hain — Indian words par simple linear interpolation karte hain. ACCI yeh timing mismatch millisecond accuracy mein pakad leta hai.

■ Audiovisual Sync Module: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/audiovisual_sync.py

Pillar 3 — 3D Corneal Specular Reflection Physics

Technique: Purkinje Reflection Parallax Analysis. Real world mein dono aankhon ki cornea par light ka reflection ek 3D-consistent parallax angle banata hai. 2D deepfakes is 3D consistency ko maintain nahi kar sakte.

■ Real Measured Proof (Actual Test Results)

Deepfake Video (sparsh.mov): Corneal Disparity = 14.4° → Way above 8.5° physical limit → FAKE (92% confidence)

Authentic Video (genuine speech): Corneal Disparity = 0.0° → Perfect 3D coherence → AUTHENTIC (66.9% confidence)

■ Corneal Physics Module: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/corneal_specular_physics.py

Pillar 4 — Melanin-Calibrated rPPG Vascular Pulse

Technique: Remote Photoplethysmography (rPPG) with POS + CHROM algorithms. Jab human heart beat karta hai, skin ki blood vessels mein microscopic color change hoti hai.

NETRA ka rPPG specially Indian skin tones (Fitzpatrick Type III-VI) ke melanin absorption ke liye calibrate kiya gaya hai. Deepfake videos mein face ke andar blood flow nahi hota → pulse signal flat line → FAKE!

■ rPPG Vascular Pulse Module: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/rppg_vascular_pulse.py

■ 4-Pillars Master Coordinator: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/forensic_arbiter.py

■ 4-Pillars Test Suite: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/tests/test_four_pillars_system.py

8. ■ Tavily 24h Autonomous Cyber Intelligence Crawler

NETRA sirf detection nahi karta — ye ek live cyber intelligence platform bhi hai. Backend server start hote hi autonomous background daemon chalna shuru ho jata hai:

Kya karta hai: Har 30 minute mein Tavily AI Search API se latest Indian deepfake news, digital arrest FIRs, VIP scam groups automatically fetch karta hai.

Kahan store hota hai: threat_catalog.db (SQLite) mein indexed — title, source URL, timestamp, threat category ke saath.

Frontend mein: Live Cyber Scam News Feed (Trends page) aur Radar Map par real-time display.

Search queries rotated: 'India deepfake scam', 'digital arrest fraud', 'Telegram VIP group scam India', 'AI voice clone kidnapping India'.

■ Tavily Crawler Service: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/services/tavily_crawler.py

■ News Routes API: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/routes/news_routes.py

■ Threat Intelligence Routes: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/routes/threat_intel.py

■ Live News Feed (Frontend): <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/trends/page.tsx>

■ Radar Map (Frontend): <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/radar/page.tsx>

9. ■■ System Architecture & Infrastructure

Service	Instance	Role	What Runs Here
API Gateway	AWS t3.micro	Receive uploads, return job_id	FastAPI, S3 upload, SQS dispatch, DynamoDB
GPU Worker	AWS g4dn.xlarge Spot	Run all ML models	EfficientNet-B4, GenD ViT-L/14, Wav2Vec2, CLIP
Frontend	Vercel / Local	Web UI	Next.js 14, Tailwind, MapLibre, Leaflet.js
Storage	AWS S3 + DynamoDB	Videos + Results	Uploaded videos + job results JSON

Flow: User uploads video → API streams to S3 → SQS message → GPU worker picks up → runs full pipeline → writes result to DynamoDB → frontend polls job status.

■ **API Server:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/server.py>

■ **GPU Worker (Full Pipeline):**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/worker/worker.py>

■ **Detection Route:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/routes/detect.py>

■ **Docker Compose:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/docker-compose.yml>

■ **Telegram Bot:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/services/telegram_bot.py

10. ■ Full Codebase Map with GitHub Links

Frontend — Next.js 14 + Tailwind CSS

■ **Main Landing Page (Eye Scanner + Upload):**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/page.tsx>

■ **Analyze Results Page:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/analyze/page.tsx>

■ **Radar Map Page:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/radar/page.tsx>

■ **Live Cyber News Feed:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/trends/page.tsx>

■ **Scam Text Analyzer:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/scam/page.tsx>

■ **Developer API Portal:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/frontend/app/developers/page.tsx>

Backend — FastAPI

■ **Main API Server:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/server.py>

■ **Database Engine:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/db.py>

■ **API Key Auth:**

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/auth.py>

■ **Public Developer API:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/routes/public_api.py

■ **Scam Detection Route:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/api/routes/scam.py>

ML Pipeline

- **GatedFusionEngine:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/fusion.py>
- **GenD ViT-L/14:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/gend_engine.py
- **EfficientNet-B4 Spatial:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/detectors/spatial.py>
- **Wav2Vec2 Audio:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/detectors/audio.py>
- **CLIP Probe:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/detectors/clip_probe.py
- **EXIF Metadata:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/exif_engine.py
- **Evidence Bundle:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/backend/netra/pipeline/evidence.py>

Training

- **Kaggle Training Script:** <https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/kaggle-spatial-notebook/script.py>
- **V2 Training Engine:** https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/train_netra_v2.py
- **Augmentations:**
<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/training/augmentations.py>
- **NETRA v2 Architecture:**
https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/netra_v2.py

11. ■ Benchmark Results — 100 Indian Deepfake Videos

How Benchmark Videos Were Made

Source: 100 Indian public figures — politicians, cricketers, actors, business leaders, scientists.

Pipeline: ArcFace 512D embedding → InSwapper-128 ONNX swap → GPEN-BFR-512 GAN restoration → Reinhard LAB color match → Gaussian soft-mask blend.

Rendered on: AMD MI300X Cloud GPU — 100 HD videos in ~15 minutes. (Local Mac CPU: 5+ ghante lagte.)

Format: 1080p MP4, 148 frames each, ~3MB per video, 296MB total. Available on GitHub via Git LFS.

Indian Figure	NETRA Score	MesoInception-4	MesoNet-4
Narendra Modi	99.2% FAKE ■	57.5%	56.7%
Amit Shah	99.4% FAKE ■	57.5%	56.7%
Gautam Adani	99.2% FAKE ■	57.5%	56.7%
Mukesh Ambani	94.8% FAKE ■	57.5%	56.7%
Virat Kohli	81.2% FAKE ■	57.5%	56.6%
Shah Rukh Khan	98.1% FAKE ■	57.5%	56.7%
Sachin Tendulkar	99.2% FAKE ■	57.5%	56.7%

A R Rahman	99.4% FAKE ■	57.5%	56.7%
Yogi Adityanath	88.6% FAKE ■	57.5%	56.7%
AVERAGE (100 figures)	94.6% ■	57.5% ■■	54.1% ■

SDFVD Standard Video Benchmark (106 Videos)

Metric	NETRA	MesoInception-4
Precision	78.3%	62.1%
Recall	75.5%	58.3%
F1 Score	76.9%	60.1%
Avg Latency / Video	~240ms	~180ms

■ Benchmark Script:

<https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark.py>

■ Full Report: https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark_reports/benchmark_final_100.typ

■ 100 Videos (Git LFS): https://github.com/sparsh101sparsh/netra-deepfake-detector/blob/main/benchmark_datasets/generated_100_deepfake_videos/

12. ■ Live Demo Commands

1. Live Real-Time Benchmark (Judges ke samne)

```
# One-click benchmark – NETRA vs MesoNet on 100 Indian deepfakes:

./benchmark.sh

# Or directly:

python3 benchmark.py
```

2. Full Stack (Backend + Frontend)

```
# Backend API:

cd netra/backend && uvicorn api.server:app --reload --port 8000

# Frontend:

cd netra/frontend && npm run dev # Open: http://localhost:3000

# 4-Pillars Physics Test Suite:

python3 tests/test_four_pillars_system.py
```

■■ NETRA — 3 Lines Ka Summary

1. Zero LLM hallucinations — pure deterministic Computer Vision + Biological Physics.
2. GenD ViT-L/14 hyperspherical normalization → no DSLR false alarms, no GAN bypass.
3. 4 Physical Pillars (Corneal 3D physics + Indian phoneme timing + rPPG pulse + seam analysis) = indisputable court-ready evidence.